

ZYVORAI LABS · CUSTOMER DOCUMENTATION

Ephemera

Getting Started

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Ephemera is a Rust control plane for disposable VMs — it needs a Linux x86_64 host with `/dev/kvm` available, and at least one of `qemu-system-x86_64`, Cloud Hypervisor, or Firecracker installed.

Prerequisites

- A Linux x86_64 host with virtualization enabled and `/dev/kvm` present.
- A Rust toolchain (`cargo`) to build from source.
- `qemu-system-x86_64` and `qemu-img` at minimum; Cloud Hypervisor and Firecracker are optional per-backend.

1. Build and install

```
cargo build --release  
sudo install -m 0755 target/release/ephemera /usr/local/bin/ephemera  
sudo install -m 0644 config.example.toml /etc/ephemera.toml
```

A one-command host bootstrap script (`scripts/bootstrap-host.sh`) installs the system packages, Cloud Hypervisor, and Firecracker for you; `scripts/deploy-remote.sh` does the same over SSH to a remote host.

2. Create your first VM

```
curl -X POST -H "Content-Type: application/json" -d '{"name": "test", "spec": {"image": "ubuntu", "memory": 1024, "cpu": 1, "disk": 10}}' http://localhost:8080/vms
```

The response is the VM's full record — id, status, disk path, PID. It boots from a disposable copy-on-write overlay of the base image named in the spec, so the base image itself is never modified.

3. Run something inside it

```
epimeta exec <id> -- echo hello
```

This runs over the vsock guest agent — no SSH, no network path required at all, as long as the guest image has the agent installed and `agent.enabled: true` was set on the create request.

4. Let it clean up on its own

Set `ttl_seconds` on the create request and Ephemera's TTL reaper deletes the VM automatically once it expires — or delete it yourself:

```
ephemera delete <id>
```

Troubleshooting

- **create fails with "base image does not exist"** — the `image` path in the spec must exist on the host running `ephemera`, and (for the `ceph-rbd` / `lvm-thin` storage backends) follow their specific reference format — see [Configuration](#).
- **exec hangs or fails** — confirm the create request set `"agent": {"enabled": true}` and the guest image actually has `ephemera-guest-agent` installed and running.
- **/dev/kvm missing** — enable virtualization in the host's BIOS/hypervisor, and confirm the current user is in the `kvm` group or run as root.

Next steps

- [Configuration](#)
 - [Common workflows](#)
 - [Admin basics](#)
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Ephemera is primarily a **CLI + REST** control plane (no first-party web console). Day-to-day work uses `ephemera` on the host, or the HTTP API the same binary serves.

CLI essentials

```
phemera create --spec examples/qemu.json
```

```
phemera list
```

```
phemera exec <id> -- echo hello
```

```
phemera delete <id>
```

REST surface

The control plane exposes HTTP endpoints for create/list/get/delete/exec and related lifecycle calls. Point clients (including Ragnarok and Zyvor Fabric) at the configured listen address.

Where to go next

Job	Doc
First VM	Getting Started
Backend & storage	Configuration
Common jobs	Workflows
Host / systemd	Admin Basics
Full topic index	PAGE_INDEX.md

Workflow	Steps
CI/sandbox VM	<code>ephemera create</code> with <code>ttl_seconds</code> set → <code>ephemera exec</code> the build/test command → let the TTL reaper delete it, or <code>ephemera delete</code> explicitly
Warm pool for fast handoff	<code>ephemera pool create</code> with a template + size → <code>ephemera pool claim</code> hands back an already-booted, paused VM in roughly resume time
Multi-host fleet	Run <code>ephemera-agent central</code> once, <code>ephemera-agent node</code> on every host, then <code>POST /fleet/vms</code> with no target node for load-aware placement
Kubernetes-native	Apply the <code>DisposableVm</code> CRD, run <code>ephemera-kube</code> per node, then create/delete VMs as ordinary Kubernetes objects
Image build	<code>ephemera build-image --spec examples/build-image.json</code> — download, verify SHA-256, resize, and customize a base image once, reuse it for every VM. See Building custom OS images for a per-distro walkthrough (apt/dnf/pacman).

For full command references, see the [technical docs](#) and the project's own [README](#). See also: [Use cases](#).